Chapter 1

Applying GenAI in Practical Scenarios

IN THIS CHAPTER

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» **Exploring creative techniques**

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This chapter covers the central touchpoints in working with GenAI models to produce unique, creative, and highly effective works. You get a feel for the tool's potential and limitations as well as advice on a few ways to overcome the inevitable hurdles that you'll encounter (which, in turn, will put you in the proper mindset to dive deeper into the details in later chapters).

In short, in this chapter, you explore the symbiosis between GenAI and your own creativity and find out how to effectively integrate AI into your work for transformative results.

GenAI as Writing Assistant

GenAI can be a wellspring of inspiration for writers and content creators. Its inspiration fuel is mined from vast amounts of data and processed for patterns humans can't see or find as quickly. That's what enables GenAI

to generate novel content ideas that might otherwise remain undiscovered. GenAI tools can suggest themes, plot developments, and stylistic elements, providing a diverse array of starting points for the creative process.

These AI-driven insights are particularly valuable whenever you face the daunting challenge of a blank page. Or in fleshing out a book idea. By analyzing existing literature and trends, GenAI can propose unique angles and perspectives, sparking your imagination and guiding you toward fresh and intriguing narratives for any form of writing or content that you wish to create. This section shows you how to use GenAI for inspiration and for writing processes.

Using GenAI to generate ideas

GenAI changes your approach to the creative process, particularly in generating ideas and drafts. GenAI models can offer you fresh perspectives, challenge conventional thinking, and encourage you to explore new ideas.

But this is not an automatic process. It is your prompts that guide the model where you need it to be of help to you. Be prepared for your work to happen both in GenAI and outside of it. For example, if you want the model to search for novel ideas and unexpected correlations, you need to prompt it to do so. You can also prompt it to provide a list of ideas on any given subject. Just be aware that vagueness in the prompt can lead to outputs that are often vague, too, and of little use to your effort. If you want GenAI to write about something specific and in a certain style, you need to include that info in a prompt.



TIP You're in the driver's seat and must steer GenAI to a specific destination, but ease up on the gas and let it suggest both direct and scenic routes for your imagination to explore. This is how you and GenAI will collaborate on ideas or finished content.

Don’t forget that GenAI images can spark ideas for you, too — even if you never use the image in the finished content. Ask DALL-E, Midjourney, Canva, or some other GenAI image model to produce an image from your imagination. The exercise may inspire your text or copy and help you shape the words you need. As they say, a picture is worth a thousand words!



REMEMBER Don’t expect GenAI outputs to ever be exactly what you want. Any true artist or wordsmith is likely okay with that. You weren’t really looking for a machine to replace you, were you? The good news is that GenAI is your assistant, not your replacement.

One other thing; you’ll need to transfer its outputs to other software to further refine it and prepare it for production — for example, as an ebook, on a website, in a content management system (CMS), or in a Word or Google doc prior to publication. Which of those you choose is up to you and also determined by the process you’ll use to bring your new creation to fruition.

Drafting content with the help of GenAI

GenAI tools can also assist in drafting a wide range of content, from business reports to creative writing. They pull information from multiple sources in their database, from which they generate coherent narratives, which can significantly speed up your drafting process.

To compose a draft with a GenAI model, follow these steps:

1. **Define your objective.**
   Clearly state the purpose of your draft. Name your goal, preferred writing style (formal versus conversational, for example), and the writing form, meaning define whether it’s for a short blog post, a video script, a long report, or a piece of fiction. Understanding the goal and nature of your finished piece will help you guide the GenAI in generating relevant content in an appropriate style and length.
2. **Select the right tool.**

Choose a GenAI tool that aligns with your stated writing needs — for instance, ChatGPT for text-based drafts or DALL-E for visual content if your draft requires imagery only. Note that ChatGPT-4o, also known as omni, will let you create images and text directly from the user interface, but some of the other versions of ChatGPT will let you pull another specialized GPT into it by using the @ symbol followed by the name of the GPT to add images and other elements as you go. See Figures 1-1 and 1-2 for examples.

3. **Craft a detailed prompt.**

   The quality of a GenAI-generated draft is often proportional to the specificity and clarity of the prompt. Develop a comprehensive prompt that includes the topic, tone, style, and any specific requirements for the draft. You can work on this outside of AI and then enter the prompt when you're satisfied that it is as good as you can make it. If you do it this way, you can cut your token usage down, which may help you cut costs and avoid bumping into usage limits. (Tokens are bits of words, spaces, characters, and the like; when added together they compose your prompt and the GenAI's response. As a rule of thumb, one token generally corresponds to around four characters or letters. Vendors predetermine GenAI usage costs per token use and daily limits by establishing a set number of tokens you can use in one 24-hour period.)

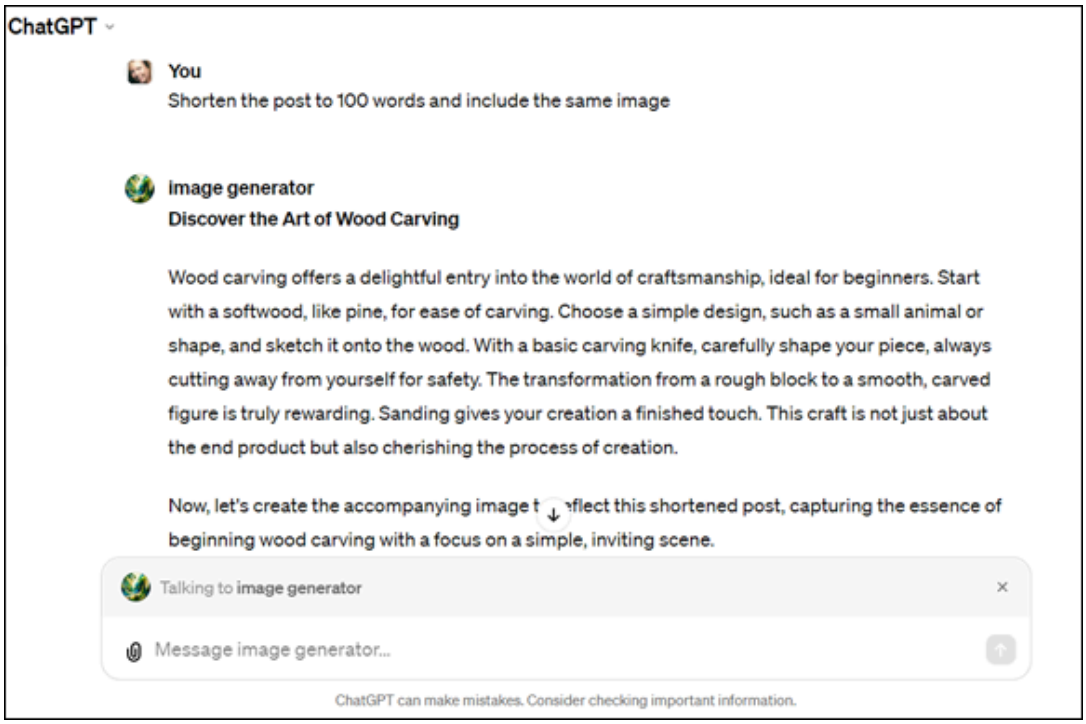
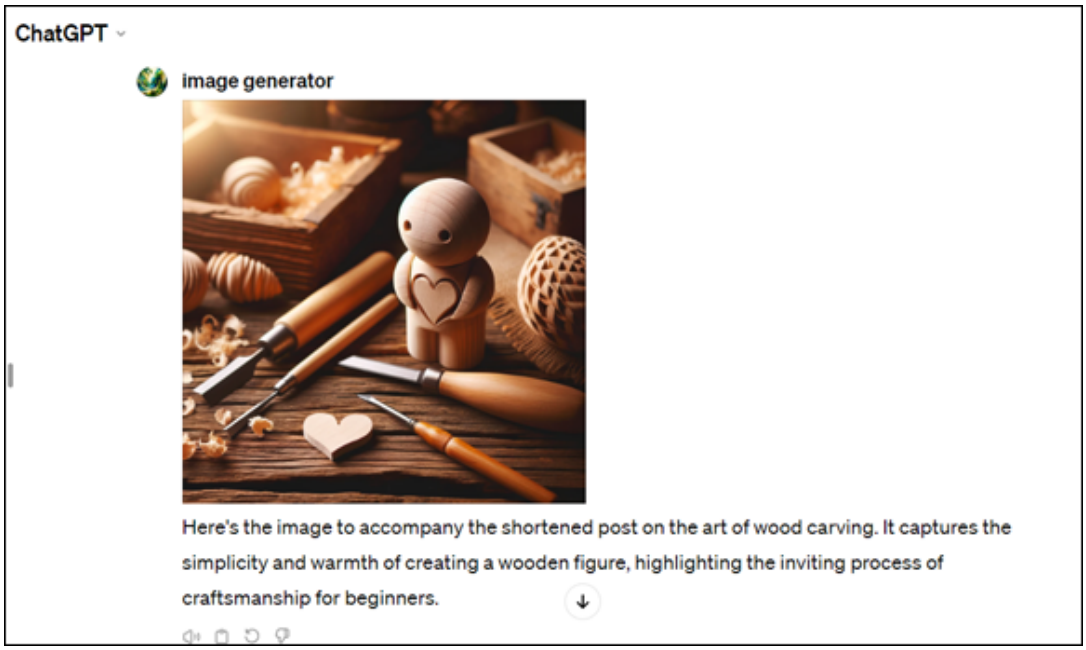


FIGURE 1-1: A screenshot of a blog post generated in ChatGPT that I'm also using @imagegenerator to create an image in the same post.



Created with the aid of Image Generator in ChatGPT

FIGURE 1-2: A screenshot of an image created by the image generator inside ChatGPT to illustrate the blog post it just wrote for me.

But you can also refine your prompts on the fly in a chain of prompts, too. In other words, what you perceive as missing or off base in the response to one prompt, you can address in a subsequent prompt, or a chain (series) of prompts as you go. This is a faster technique than thoughtfully composing prompts off-line, but you run the potential risk of upping your token counts.

4. **Input the prompt.**
   Enter the prompt in the prompt bar on the GenAI tool user interface. Some models allow you to attach images and other files to the prompt, but you still need to type instructions for the model in the prompt bar, too.
5. **Review the output.**
   Examine the initial draft generated by the AI. Look for relevance, coherence, and alignment with your original objective. Also, check for accuracy and plagiarism, too.
6. **Refine and iterate.**
   You can further its improvement by tweaking the language and adding more context in the next prompt or by adjusting the parameters of the task for the model in a system message, retrieval augmented generation (RAG), or, to a lesser degree, in a prompt.
7. **Edit and personalize.**
   When a satisfactory draft is generated, edit the content to add a personal touch, ensuring it meets your standards and reflects your voice. Double-check again for accuracy and plagiarism.
8. **Utilize additional tools.**

If necessary, use other GenAI tools for specific tasks, such as specialized GPTs like those you'll find in ChatGPT Plus versions and higher. Refer to Figures 1-1 and 1-2 for examples. But you can use other types of GenAI tools, too, such as new search tools like Perplexity.



TIP GenAI models are trained on a dataset, which means that data has a cutoff date and can age out (a process called *model drift*). To overcome this limitation, use a supercharged, GenAI-enhanced/embedded search engine like Perplexity, Google Smart Search, or Bing to get more current information. Then copy that response and feed it into ChatGPT prompt bar with instructions to include the current information in the draft! This process is a type of AI chaining, meaning you're using the output of one AI as part or all of a prompt in another AI.

9. **Finalize the draft.**
   Make the final adjustments to the draft, ensuring it is polished and ready for its intended audience or purpose. Typically, GenAI-generated text sounds like it's machine generated. Make copy changes as needed to refine its style to fit your intended audience.
10. **Save and archive.**
    Store the final draft and any notes about the process in an organized manner. This practice can be useful for future reference or for improving subsequent interactions with GenAI tools. You'll want to make a prompt library or prompt archive, too. This way, you can readily reuse a successful prompt when needed and, thus, save time and effort in recreating the same prompts later.

Sprucing up your writing with GenAI

GenAI offers writers an array of tools to enhance creativity in their writing. By integrating the following tips with GenAI tools, you can push the boundaries of your personal writing style, producing innovative and engaging content:

- **Include elements like genre, tone, and style in your prompts to get tailored results.** You can even provide an example of writing that you want GenAI to follow or mimic. For example, provide it with a text sample from an academic paper in the prompt and instruct it to use a similar tone and vocabulary. However, don't be surprised if it refuses to mimic writing samples from current writers or even your own writing samples, as many GenAI model builders have built guardrails against GenAI copying such work in an effort to prevent the model from plagiarizing or committing a copyright infringement.

- **Consider prompting the model to suggest expansions, twists, or alternative directions that a competitor may produce to compete with your work.** This way you can capture that information in your piece or produce another work from that angle ahead of your competitor.
- **Experiment with GenAI to get advice or additional perspectives from celebrities, historical figures, or famous authors or to blend different writing styles, which can lead to unique and creative narratives.** For example, you can prompt GenAI to produce critiques or suggestions on your content from Einstein, Dr. Seuss, Margaret Thatcher, President Lincoln, Steven Spielberg, and Stephen King. What an insightful conversation that will be! But you're not limited to these people. Create your own board of creative advisors!
- **Use GenAI for editing and refinement.** The AI can offer rephrasing suggestions and word alternatives that may enhance your original draft. You can also prompt it to expand, summarize, make lists, or whatever you need to further refine your draft.

TURNING ORDINARY INSTRUCTIONS INTO MEMORABLE CONNECTIONS

Not all content in this world is interesting. Take, for example, the badly drawn and often mystifying laundry instructions on a clothing label, assembly instructions for a piece of furniture, or even some patient instructions for everything from medicines to hospital discharge papers.

When instructions are boring or confusing, people don't bother. When they don't bother to read and follow them, mistakes happen, and bad outcomes can follow. A little creativity can be a profound fix for those problems.

For example, if you're writing patient information for a child to follow, you can use GenAI to create it in the style of Winnie-the-Pooh, so the child finds it easy to understand and follow without feeling fear or intimidation. It might look something like this:

- **PROMPT:** Write patient information for a young child (5 to 10 years old), reminding them not to get their bandage wet. Make it short, memorable, and in the style of Winnie-the-Pooh.
- **RESPONSE:** *In the Hundred Acre Wood, where the streams babble clear, there's a small, simple thing that Pooh Bear holds dear. "If you have a bandage," Pooh says with a smile, "Keep it away from the water, just for a while. Whether it's bathtime or puddles or raindrops that fall, keep your bandage quite dry, away from it all. For a bandage that's dry is a bandage that's true, and it helps you to heal," says*

Winnie the Pooh. So hum a small hum, think of honey, think sweet, Imagine your bandage staying dry as your feet. And soon, when it's off, you'll be happy to see you're as right as rain, as healthy as can be!

Getting a Visual Assist from GenAI

Making use of GenAI can be a transformative experience for artists, designers, and content creators. For content creators, GenAI can create images to illustrate the content, attract readers, and increase reader retention rates. For visual creators, be they fine artists, fashion designers, graphic artists, or product designers, creating visuals with the help of GenAI is the entire point of the exercise.

Fortunately, GenAI is accessible to anyone to create any image. Whether that image is what you want — or results in commercial success — depends on the level of your own creative talent.

GenAI in graphic design and visual arts

GenAI has become an indispensable tool in graphic design and visual arts, offering artists and designers ways to expand their creative horizons and streamline their workflows. Creatives can now generate complex patterns, simulate intricate textures, and even create entire compositions with a level of detail and speed previously unattainable.

GenAI models like DALL-E, Midjourney, Canva, Adobe Firefly, and Stable Diffusion enable artists to bring complex visual ideas to life from simple text prompts. These models are trained on extensive datasets of images and their descriptions, learning to generate new images that are rich in detail and creativity. The models can also create images in virtual space, unconstrained by physical limits. They can render images with a mix of textures or mediums that may not have been considered or possible in physical space.

GenAI also enables the manipulation of images, such as changing backgrounds and adding elements to photographs. GenAI is embedded in software like Adobe AI programs that enable image creation and seamless manipulations in previously unprecedented ways. You can even create

royalty-free custom “stock photos” with GenAI since it can create human characters who do not actually exist and, therefore, require no permissions or payments.

While GenAI incorporates elements of computer vision, it enhances the design process through pattern recognition across vast datasets, predicting elements that will engage audiences.

Generating visual content with AI tools

If you’re looking to create images for your content, try entering part or all of the content in the prompt bar and prompting the model to provide illustrations for key points. Odds are that you’ll be delighted at the results. You can also prompt the model to generate an image according to your description in the prompt and customize that image in subsequent prompts until you have exactly what you were looking for!

Even budding artists are spinning text into visual gold. GenAI-created images can be found for sale in a multitude of marketplaces, from artist- and gallery-owned websites to Etsy and Adobe stock images websites.

Here are a few tips on how to tease some great images out of GenAI models:

- **Get specific with your prompts.** It’s all about the details. Dictate in the prompt what you see in your inner creative vision. Include details like lighting, shadowing, perspective, character traits, apparel, the scene or backdrop, a time period, a medium (maybe watercolor? something else?), the general mood, and other information. Imagine you are moving the GenAI like you would an artist’s brush or a photographer’s camera. *If* you do so skillfully, the resulting image will be amazing!
- **Create in layers.** Write your prompt in descriptive layers. Describe specific elements in layers so you don’t forget to include important details. Try stacking different themes, eras, stances, moods, and actions to create complex, eye-catching visuals.

  Use GenAI to adapt your sketch or an image. You can add images to your prompts in several models. Have you already done a sketch? Put it in the prompt and ask GenAI to finish it as a painting, a photograph, or even a sculpture. You can also put an image in the prompt and instruct the model to extend

the scene beyond the frame or the borders of the image with new elements. In DALL-E this is called *outpainting*.

- **Polish with iteration.** Start with a rough sketch from the AI and then refine, refine, refine. Nudge the AI to tweak colors, shuffle the composition, or add a little something-something until it's just right.
- **Mix and match styles and mediums.** Guide the AI to fuse different artistic styles, periods, or mediums into a fresh, new visual work. Remember that GenAI has no physical limits, really, so go ahead and see what your talent can create when there are no limits.
- **Embrace happy accidents.** Sometimes the best ideas come from a "whoops" moment. Let the AI surprise you, and you may just like that image better than the one you initially had in mind.
- **Draw from everywhere.** An image may be worth a thousand words, but its value is a thousand more when there's a story behind it. Don't just stick to art lingo when writing your prompt. Borrow from science, tech, or your favorite novel to prompt the model to give your images a story that's as deep as they are stunning.



REMEMBER Despite the technological advancements, the artist's intuition and talent is vital in guiding GenAI to produce work that resonates on a human level. As GenAI evolves, its capabilities will balloon, providing artists with an expansive canvas for innovation and exploration.

Harnessing GenAI for even more visual creativity

If you're a professional image creator, you'll likely want to exert more of your talent in the GenAI's image generation. Following are some tips for fine artists/video artists, graphic design/visual communications professionals:

- **Experiment with AI-generated imagery.** Use GenAI tools to create new visual concepts. Input descriptive text to prompt a model like DALL-E, Midjourney, or Stable Diffusion to generate images or patterns that can serve as a starting point for your creative projects. This can be particularly useful when you're looking for fresh ideas or want to explore how AI interprets your descriptions.
- **Play with perspectives.** Use GenAI to show you your image from a different perspective, or even from more than one. This can spur new ideas or ideas on how you might want to change the image you're already working on. You can

also use GenAI to extend your image past the frame, meaning past the edges of your design.

- **Blend human intuition with machine precision.** While GenAI can generate a vast array of visual data, human intuition is key to selecting and refining these outputs. Use your judgment to curate and modify AI-generated images, combining them with traditional artistic techniques to create unique and compelling visuals. This blend of human and machine can lead to innovative designs that may not be possible through conventional methods alone.
- **Leverage AI for enhanced pattern recognition.** GenAI can identify and replicate complex patterns, which can be used to inspire new artworks or designs. For example, prompt the model to compare your newly created image with modern trends in advertising or commercially successful but competing product designs. The responses may spur you to create a more successful ad or product.

Problem-Solving with AI in Creative Projects

If and when you find a problem in your own creative idea and can't find a solution, GenAI can help with that. Perhaps something is off in your painting, either technically or creatively. Take a picture of it and add that to a prompt to an image generator, asking it to analyze and offer suggestions to solve your stated problem or improve it overall. You can do the same with your songwriting, your ad campaign, your novel, your movie script, or any other creative work.

GenAI can quickly show you any issues, suggest solutions, or guide you to another way to express your creative thoughts that you may like better. Even creators who shy away from showing their work in its early stages to other people find it easy to privately share it with AI.

But sometimes the AI *is* the problem.

If you're like most serious creators using GenAI to create unique and professional-level works, you're bound to be frustrated out of your mind at some point. Don't worry, that's normal.

The good news is that you can turn GenAI models around. Yes, the model is the source of the problem, but it can also come up with the answer or at least a workaround. Here are a few tips and tricks on that and some other things you can do to get unflustered and back on schedule.

- **Change up your prompts to redirect the model.** Start a new chat in either the model you're currently using or a different model that's text-based and perhaps a little better suited to problem-solving. Remember, oodles of models on the market are specialized in different types of work. If you have the time, or the problem is vexing enough to warrant the exercise, go find one and get it to solve the issue. If you don't want to go to all that bother, prompt the nearest GenAI model to give you some problem-solving options.
- **Try again or give the same command in another way.** If you're trying to get the model to do something specific and it seems to be doing everything BUT THAT, tell it in another way. I promise, the GenAI is not trying to drive you crazy. It just doesn't understand what you want it to do. Remember that it doesn't actually understand anything. The model is completing a pattern from the pieces you gave it in the prompt. So, give it different pieces of the puzzle to work from. Restate your prompt from one or more other perspectives or ways, and you'll finally get some movement toward the response you desire.
- **Form a digital committee.** I know, I know, everyone hates to deal with committees when they're seeking a definitive answer to something. But this is a digital committee wherein you are prompting the GenAI to assume several professional or community roles simultaneously in order to offer you many different possible solutions in a single response. Take what you want and delegate the committee to chat history forevermore. Don't you wish real committees were that easy to dismiss?
- **Restart the GenAI application.** Yes, I know, this sounds a lot like "unplug it and plug it in again" advice from ages past. But it's true. For any number of reasons, or none at all, a GenAI app can freeze, become lazy, or otherwise get hung up on error messages. Save yourself and your IT support some grief and just shut it down and restart.
- **Are you sure you prompted what you actually wanted?** People (you and me included) tend to shorthand our thoughts, and that can carry over in our prompts. While another person, especially someone who knows you well, can deduce what you meant even though that isn't what you said, GenAI is too literal to do that. So, if you aren't getting the response you need, back up and rethink your prompt. Did you ask what you meant to ask? You might want to reword that prompt so it is clearer.
- **Use the prompt to stop repetitive phrases.** Is the GenAI model using repetitive phrases or words? Forbid it to use those phrases or words in your prompts or the system message. Voila! Now, the issue is fixed. (At some point, you might want to unfix that, though, as it may genuinely need those phrases and words back to legitimately answer a prompt down the road.)
- **Layer AI to stop repetitive responses.** If the model keeps giving you the same responses simply reworded, check your prompts to make sure you aren't asking it in different ways to do the same thing. If prompting isn't the issue, add data

for the model to use. That's quite often a pain to do, however. An easier workaround is to add data in your next prompt. One way to do that is to prompt another GenAI model with the same information and feed its response into your next prompt in the GenAI application or model you were initially using. This is a form of AI chaining that simply means using all or part of an output from one AI model as part of the prompt in another. Another way to use multiple AI models is to aggregate them by using different models to perform specific steps necessary to complete a larger task. For example, you might want to use one GenAI model to generate text, another to create images, and a third that's embedded in other software to publish the two outputs in a single format such as a document.

- **Consider something completely different.** In the realms of architecture, fashion, and graphic design, GenAI can weave together designs that defy convention, crafting spaces, garments, and visuals that belong in the galleries of tomorrow. In other words, if the way you are currently pursuing is not working, then prompt the AI to take an entirely different artistic route to the same end.